

BOSTON UNIVERSITY
ELECTRICAL & COMPUTER ENGINEERING

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EC601

*advanced research,
real products, led by you*

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Graduate level Thesis, Project or
Company Lead R&D Project

From polymaths to subspecialists — and then?

What we know — click the sources

One scholar, many fields. Aristotle: logic, physics, biology, ethics, poetics. Ibn Sina: medicine, astronomy, philosophy.

[\[source\]](#)

1884 — "electrical engineer" becomes a profession.

AIEE founded; MIT began teaching EE in 1882. [\[source\]](#)

1963 — AIEE + IRE merge into IEEE. One society then; 39 technical societies today. [\[source\]](#)

Medicine, same curve. 24 boards, 40 specialties, 89 subspecialties (ABMS). [\[source\]](#)

1991 — arXiv starts as one physics list. ~150 subject categories today. [\[source\]](#)

What happens next — hypotheses

A. Specialization continues. Every knowledge technology so far (print, computing, the web) increased it.

B. Bimodal split. Frontier specialists go deeper; AI-augmented generalists integrate across domains.

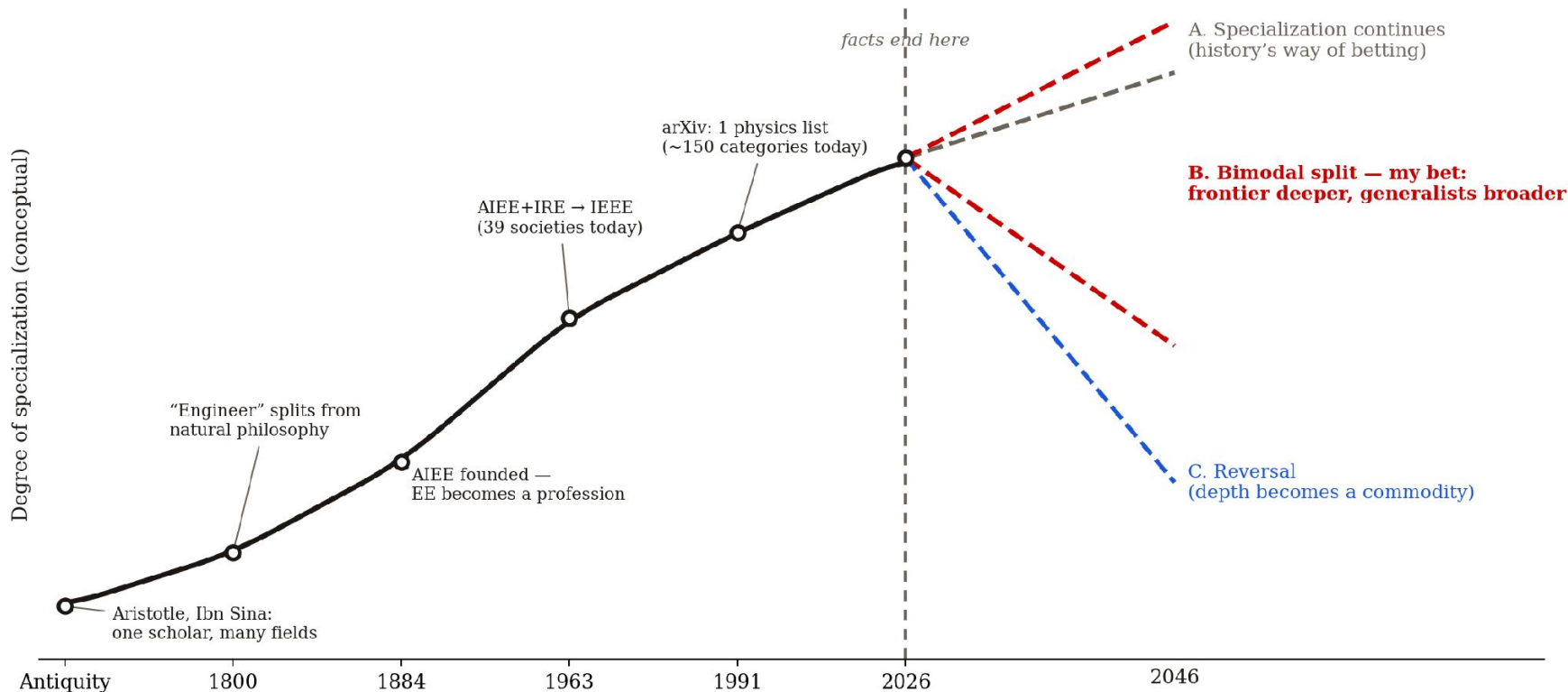
C. Reversal. Depth becomes a commodity; breadth wins broadly.

My bet: B. *Previous tools cheapened access to knowledge; AI cheapens the execution of procedural depth itself — so human advantage shifts to integration across domains. Debate me.*

2026 — facts end here

Sprint 0: which scenario do you believe — and what evidence would change your mind?

One picture, three futures: where do you place your bet?



Solid curve and dashed futures are illustrative. The dated points are documented facts — sources on the previous slide. Time axis not to scale.

Why I believe in the split — argue with me

1. Companies want more from one person

One individual orchestrating many AI agents. Meta reports engineer output up 30% since 2025 (power users +80%) [\[Axios\]](#); Zuckerberg: projects that took big teams now done by one talented person; 7,000 staff moved into an Agent Transformation group [\[Fox\]](#).

The counterpoint is real: Meta's agent-replacement plan stumbled — code changes +220%, user-visible features only +36% [\[Reuters\]](#). Orchestration is a skill — that gap is exactly what this course teaches.

2. AI as the professor's PhD student

In 2026, an Erdős problem (#728) was solved “more or less autonomously” by AI — Terence Tao's words [\[Tao\]](#). Tao called an earlier LLM solution of #281 “perhaps the most unambiguous instance” of AI solving an open problem; DeepMind's AlphaProof produced Lean-verified proofs of nine more [\[tracker\]](#).

Tomorrow's PhD student directs the research capacity of yesterday's professor. Collaborative research needs the same orchestration skill — plus the depth to verify.

3. A single skill cannot sustain itself

The mechanism from the previous slides: earlier tools cheapened access to knowledge; AI cheapens the execution of procedural depth itself.

Whatever single procedural skill you master, the price of executing it is falling. What does not fall: knowing what to build, for whom, and whether the output is right.

So advantage shifts to integration across domains — with enough depth in one to tell good work from confident nonsense.

You have one advantage no one can take: you're starting now, with these tools, without old habits to unlearn. Don't waste it on being “good with AI” — everyone will be. Go deep enough in one thing to catch the machine's mistakes, and wide enough to connect what others keep apart.

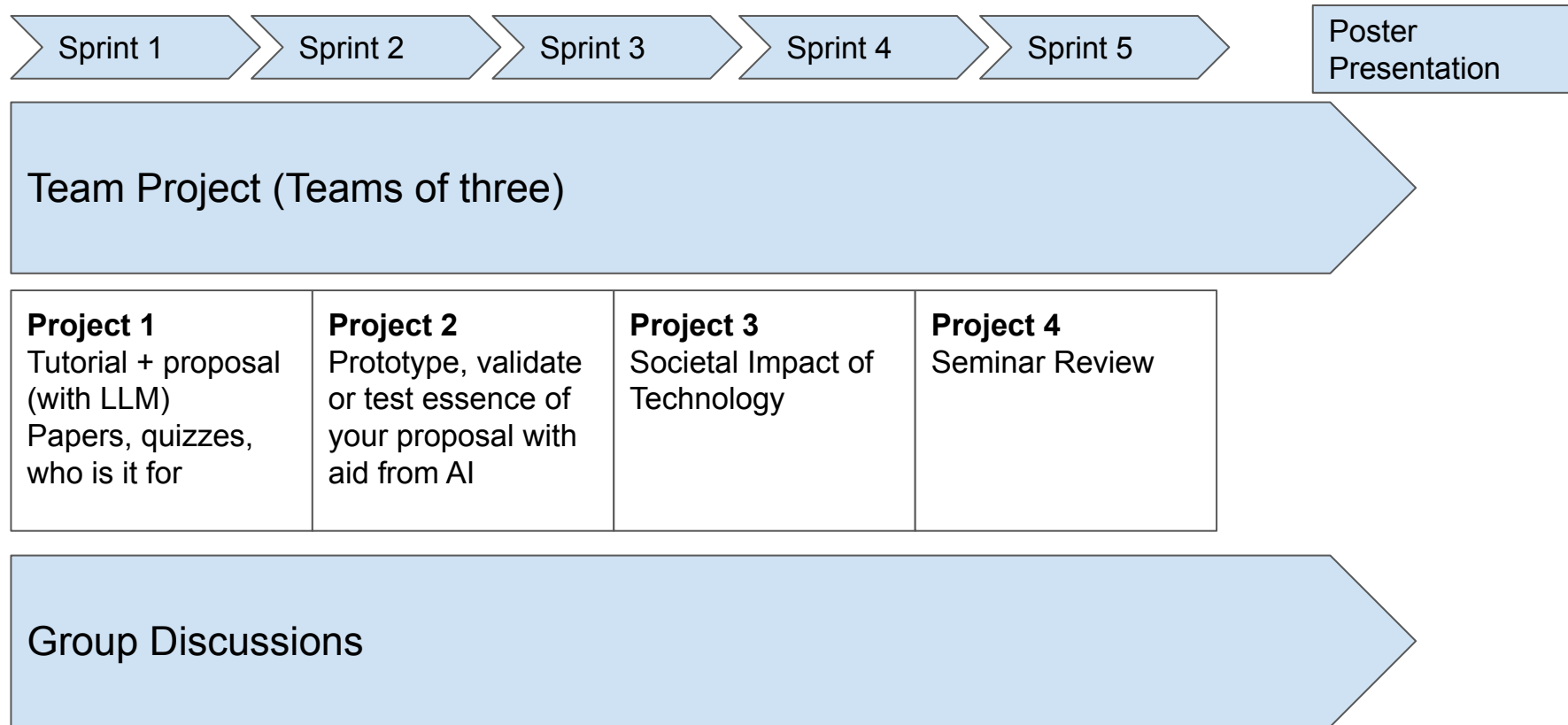
Preparing you to be a technology leader

- **Product formation:** how to take an idea and make it into a product
- **Product analysis:** competitive analysis, landscape analysis
- **Use case development:** translating use cases into system and engineering requirements, resulting in a system architecture
- **Product testing:** how to test your product
- **Concept validation:** how to validate concepts
- **After launch:** maintain, monitor and upgrade products

Class Style

- Self-learning
 - You need to research, read and train yourself with help from us.
- Utilize AI as a companion and a colleague BUT you are responsible for the outcome
- Agile Learning and development
- Class collaboration
- Very interactive and you need to present
- Always have your computer with you in class

Class Plan



Academic Misconduct (copied from Prof. Clem's email)

BU takes academic integrity very seriously. Academic misconduct is conduct by which a student misrepresents his or her academic accomplishments, or impedes other students' opportunities of being judged fairly for their academic work.

Knowingly allowing others to represent your work as their own is as serious an offense as submitting another's work as your own. More information on BU's Academic Conduct Code, with examples, may be found at

<http://www.bu.edu/academics/policies/academic-conduct-code>

Collaboration Policy (adapted from Prof. Clem)

- This is a research and project-based class. You may build on open-source code and published research — but you must reference it, explain why you used it, and follow its license. If the license does not allow use, do not use it.
- You may talk to people outside the class (engineers, researchers, potential users). Credit them as sources, just as you would a paper.
- All reports and papers must be written in your own words. Code may be shared within your team.
- Acknowledge every source — papers, code, people, and LLM chats.
- Failure to do so constitutes plagiarism and will be treated as cheating. Not sure whether something is allowed? ASK ME first — it is much easier than explaining it afterwards to the college disciplinary committee.

LLM

- You are expected to utilize AI LLM during the whole class.
- LLM is a companion or someone who works for you. This means you are responsible for the outcome
- You MUST validate every interaction
- You must include the reference to the chat in your reports
- You must include instruction to the LLM to
 - Not to make up facts
 - Not to just agree with you but to debate
 - Provide references from refereed and/or trustworthy sources
- When you are learning a new topic using AI, you must ask the LLM
 - Provide quizzes during the interaction to validate your understanding of the topic

How Classes Work

- You are expected to work on your own. Some classes will be cancelled in lieu of in person meetings with you.
- In class will be for 1 hour
- We will have open office hours for support
- We will have a teaching assistant assigned for groups
- Introduce the ambassadors

All online: I want it to be part of your resume

- All your homeworks, documents and projects must be on GitHub or shared drive
- Your GitHub account is required
- The next week, focus on becoming familiar with Github and principles of code management and agile development

Grading

A note on grading. This rubric is a guideline, and its real purpose is feedback — it tells you where your work stands and what to fix. There is no curve and no quota: do the work properly, revise when I push back, and you will get the A. The only way to lose it is to not go deep

Item	Item	Grade	Totals	Details	
Team Grade	Sprints	20%	40%	Main criteria is effort during the sprint period.	
	Final Deliverable	20%		Main criteria is poster presentation, your overall effort during the project	
Individual	Project 1: Proposal	15%	60%	Main criteria is covering the main areas the report requires	
	Project 2: Prototype	15%		Will be defined during user stories lecture	
	Project 3: Societal Paper	10%		Will be defined during societal impact lecture. Part of the grade is for group presentation.	
	Project 4: Seminar Review	10%		Pick any scientific seminar, attend it and do a thorough literature review of it.	
	Participation	10%		Attendance, in-class discussions and presentations, and engagement in the 1:1 team meetings that replace some lectures	
Total			100%		

Seminar Review

- Adapted from Prof. Pisano's Senior Design course.
- Objective: literature review of a topic and understand it well. Like Project 1, but focused on an area where people have already worked and achieved results.
- Expectations:
 - Pick a seminar, presentation, PhD defense, etc. at BU or outside BU and attend it
 - Read the related papers and work
 - Write a paper summary describing
 - Topic
 - Main contributions
 - Results
 - Your view of the work
 - What you would do as next steps
- Due date: submit before December 1, 2026
- Resources
 - BU ENG events
 - BU CISE events
 - BU College of Computing and Data Sciences events
 - MIT Media Lab events

First Project

- Objective: Pick an area of research, topic, investigate it and put a proposal for an idea you want to pursue.
- [EC601 — Phase 1: Research & Project Areas](#)
- [The assignment description](#)
- [Survey](#)
- Create a Folder with in [The submission folder](#). The folder should be named: FirstName_LastName

Next week plan

- Wednesday 9/9/2026:
 - I will group you based on your topics
 - First 45 minutes: You will exchange ideas in groups
 - Second 45 minutes: Few groups will present their ideas
- Project 1 is due 9/12/2026

Questions

Thank you